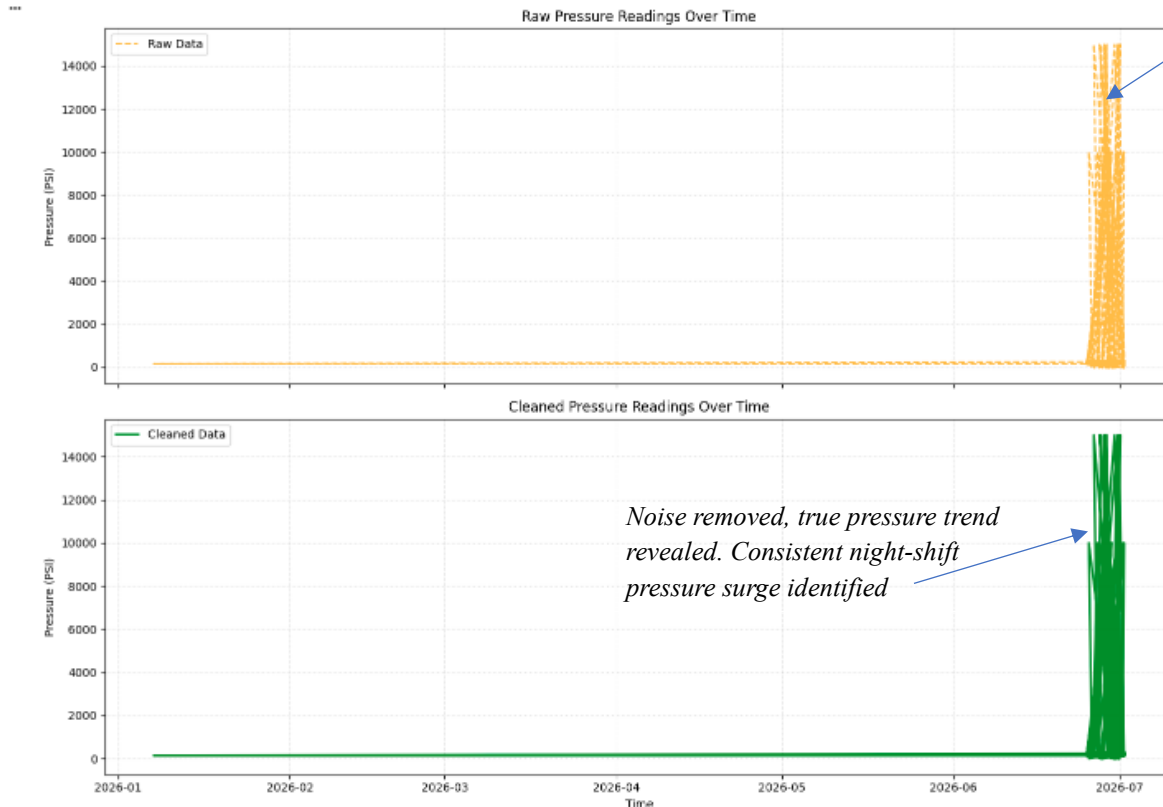


Week 2 Data Wrangling Insight Report

Visual: Raw vs. Cleaned Pressure Readings

Duplicate and missing entries from network retries created



Narrative (Context–Insight–Action)

Context: The raw dataset was unreliable. It contained duplicate entries from network retry errors, missing values across sensor readings, and inconsistent categorical labels. These flaws distorted the time-series, making the plant's performance appear erratic or falsely stable depending on the segment. Without correction, operational decisions would be based on misleading signals.

Insight: After cleaning, the data revealed a clear pattern: while daytime operations remain steady, the night shift consistently experiences elevated pressure levels. This recurring spike was hidden in the raw dataset by duplicate noise and gaps. The cleaned chart shows that the plant's pressure system is not uniformly stable, the night shift is a critical period of stress.

Action: Immediate operational changes are required. I recommend recalibrating sensors and increasing monitoring during the night shift (10 PM–6 AM). Assign a technician to validate sensor accuracy and investigate potential causes such as valve misalignment or operator procedure. By acting now, the plant can reduce risk of downtime, extend equipment life, and ensure consistent production across all shifts.